Host Utilities

TATA

TATA POWER

TATA POWER-DDL
Towards a Greener Tomorrow

BSES
BSES Rajdhani Power Limited

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ORGANIZER

ISGF
India Smart Grid Forum

India
SMART UTILITY
Week 2026
10 – 14 March 2026, New Delhi

Supporting Ministries

सत्यमेव जयते
MINISTRY OF POWER
GOVERNMENT OF INDIA

सत्यमेव जयते
MINISTRY OF POWER
GOVERNMENT OF INDIA
CENTRAL ELECTRICITY AUTHORITY

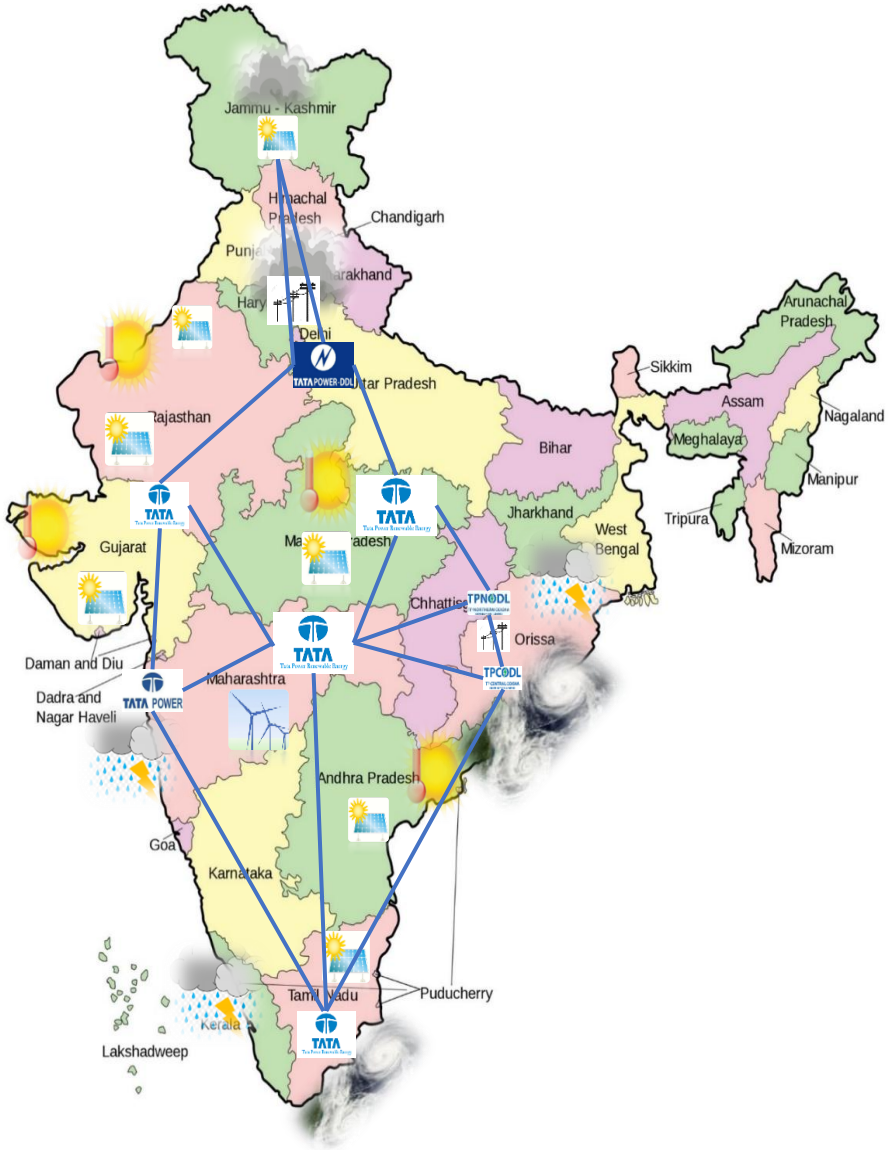
Session : ISUW 2026 Technical Paper Presentation Session

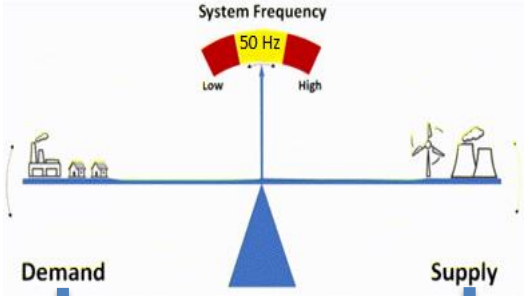
PRESENTATION TOPIC – WeatherXpert : The PAN Tata Weather Platform (26098)

Presented By

Ms. TRUSHA BISWAS – HEAD TECHNOLOGY & QUALITY

INTRODUCTION





Low Forecast Accuracy

Distribution
1. Load Forecast
2. Major Outages
3. Load Shedding
4. Penalty

Generation
1. Generation Forecast Accuracy
2. Outages
3. Damage

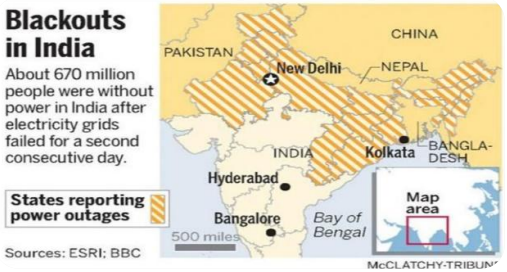
High Forecast Accuracy

High Grid Stability

Overburdened Tie Lines

Low Grid Stability

Power Outages



Financial Penalties

Accurate forecasts prevent financial losses from grid

DEVIATION SETTLEMENT MECHANISM

CERC Regulation

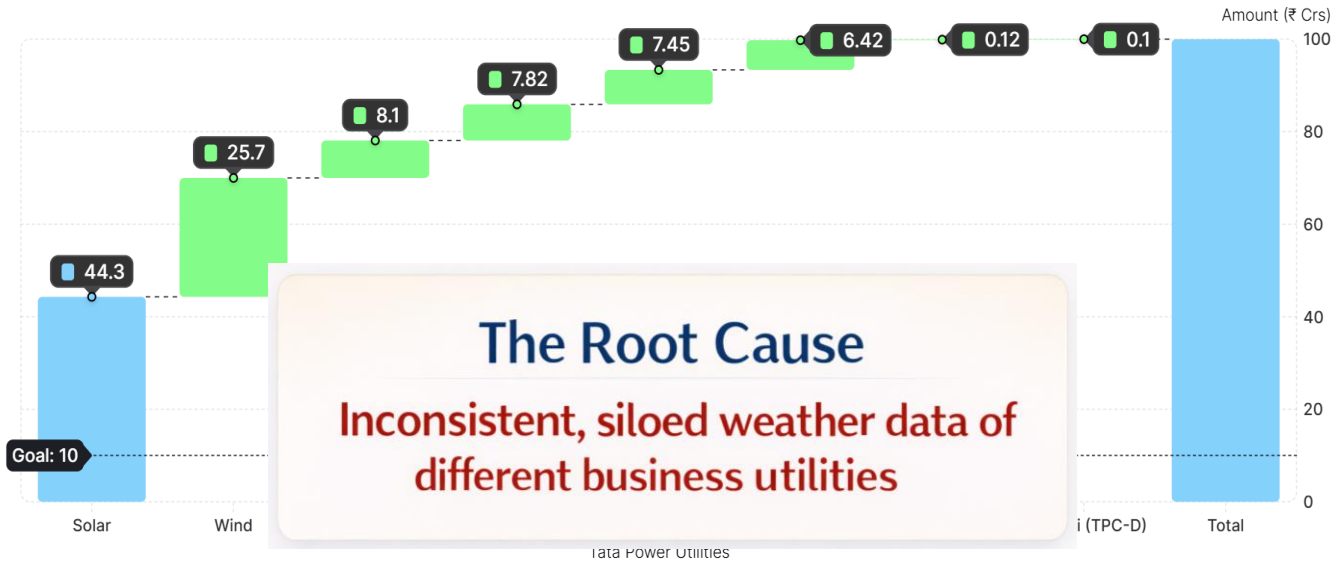
Deviation Band
- Demand = +/- 1%
- Non RE Supply = +/- 5%
- RE Supply = +/- 10%

CONTEXT

Weather Impact Areas on Power Utilities

- Grid reliability affected by unpredictable weather patterns
- Load forecasting inaccuracies lead to resource misallocation
- Renewable generation planning hampered by forecast limitations
- Operational resilience compromised during extreme weather

UTILITIES-WISE PENALTIES
Distribution & Generation



The Root Cause

Inconsistent, siloed weather data of different business utilities

Ready made Solutions

TATA POWER → BluWave-ai ✗

TATA Tata Power Renewable Energy → MITSUBISHI ✗

WEATHER SERVICE PROVIDERS

AccuWeather ✗ | skymet weather wise ✗ | ✗ | ✗

DATA NOT AVAILABLE IN REQUIRED GRANULARITY

- 1 Min ✗
- 15 Min ✗

- Intraday ✗
- Day Ahead ✗
- 15 Day Ahead ✗

PRICING PRICING PRICING

Data Acquisition and Preprocessing

Data Sources

IMD Grid Weather	GFS/ECMWF NWP Models	Satellite Cloud Imagery	AWS Ground Stations	SCADA Telemetry
1–3 hr	1 hr	10–15 min	1 min	10 sec–1 min

Data Challenges

- Noise and Spikes in GHI DNI DHI
- Timestamp Misalignment across sensors
- Cloud – induced fluctuations distorting Data Curve

Data Cleaning & Alignment Techniques

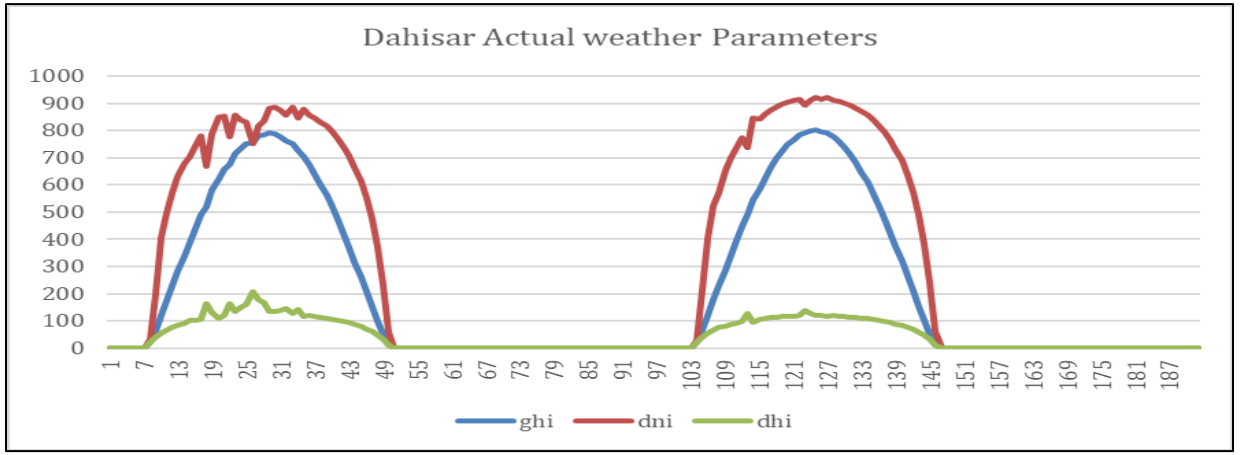
- Smoothening Centered Moving Average
- Outlier Removal Threshold based replacement
- Alignment Cross co-relation to correct sensor phase

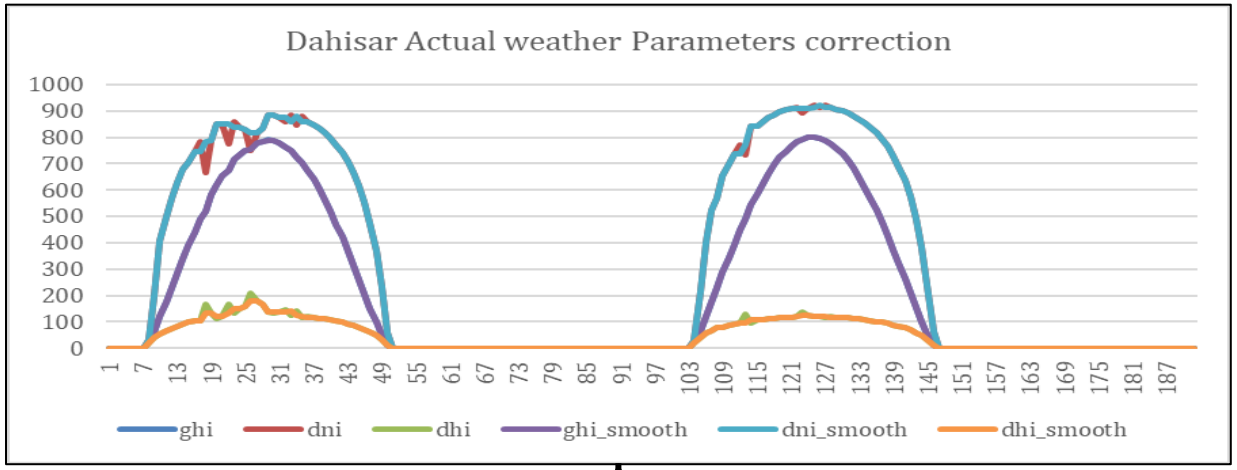
$$\hat{x}_t = \frac{1}{k}\sum_{i=-m}^{m} x_{t+i}, \quad k = 2m+1$$

$$X_{aligned}(t) = X_{source}(t - \delta t_{delay}) + \Delta_{bias}$$

F(x) used for the corrections

Enables High Accuracy and Robust SCADA Performance Diagnostics





Output

- Physically Realistic Solar Curves
- Consistent GHI/DNI/DHI Alignment
- Clean Data Set ready for Forecasting & Performance Analytics

Layered Methodology

Data Integration

Data Ingestion

- Each source is getting ingested through ETL Pipelines using REST API FTP and Feeding Protocols

Data Cleaning

- Noise Reduction
- Outlier Detection
- Cloud Impact Correction

Temporal Alignment

- Due to varying resolution (10Sec to 3hrs), all stream resume to common time base.

$$X_t^{(\text{aligned})} = \text{Resample!}\left(X^{(\text{raw})}, \Delta t\right)$$

Data Fusion

- A Hierarchal fusion model combines forecast Data (NWP) Real Time Obs. AWS SCADA and satellite Imagery

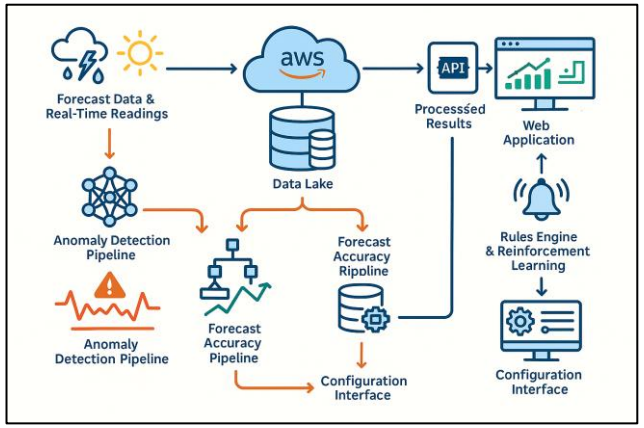
$$X_t^{(\text{fused})} = w_1 X_t^{(\text{AWS})} + w_2 X_t^{(\text{NWP})} + w_3 X_t^{(\text{Satellite})}$$

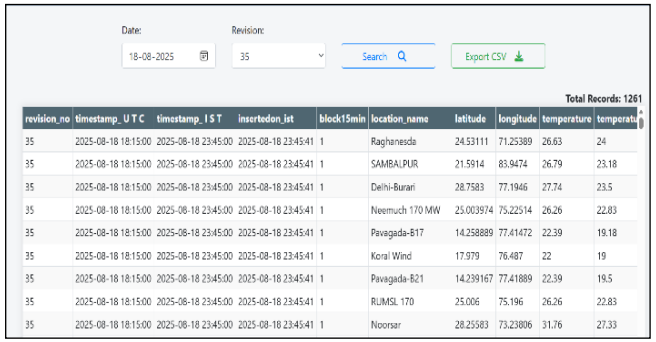
System Integration

- Real Time Dashboard
- API endpoints
- Validation viz SCADA Telemetry

AI-ML Calibration

The system aggregates multi-source weather forecasts and real-time data from over 100 Tata Power stations into an AWS data lake, where it is standardized and processed. Two ML pipelines then detect anomalies and continuously improve forecast accuracy using neural networks and ensemble models. The refined outputs are delivered via APIs, a cloud web app, and an intelligent alerting system, all configurable through a scenario-based analytics interface





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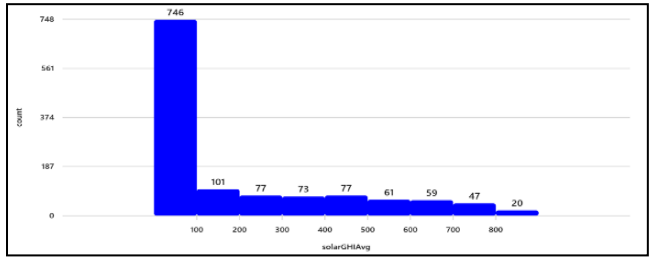
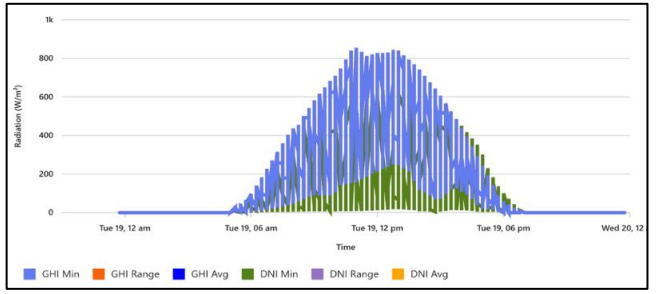
Probabilistic Forecasting

- In this study, uncertainty intervals were generated using historical and forecasted irradiance data. For each time block, percentiles (5th, 50th, 95th) were computed to represent pessimistic, most-likely, and optimistic scenarios.
- Strong variability in irradiance was observed: GHI: P95 ≈ 708 W/m²DNI: P95 ≈ 501 W/m²Low median values indicate many non-sunlight blocks.

$$P(q) = Quantile(Xt, q), q \in 0.05, 0.50, 0.95$$

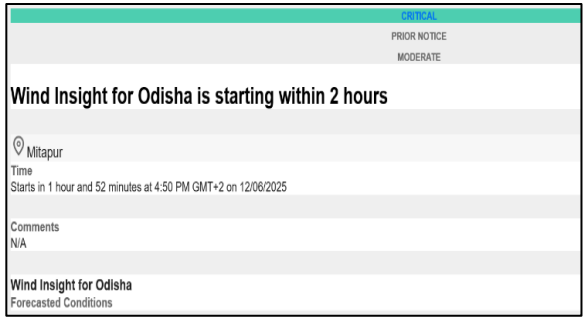
Metric	GHI (W/m²)	DNI (W/m²)
Mean	178.6	99.8
Std Dev	244.9	162.9
Min	0	0
Median	19.2	5.73
95th %ile	708.6	501.1
Max	855.2	626.8

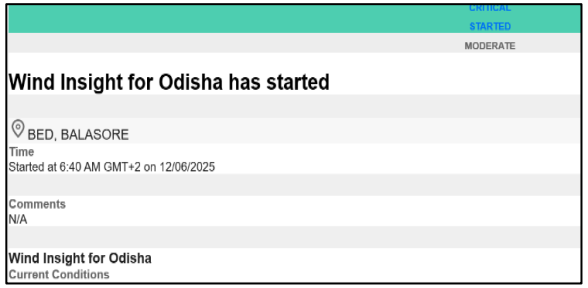




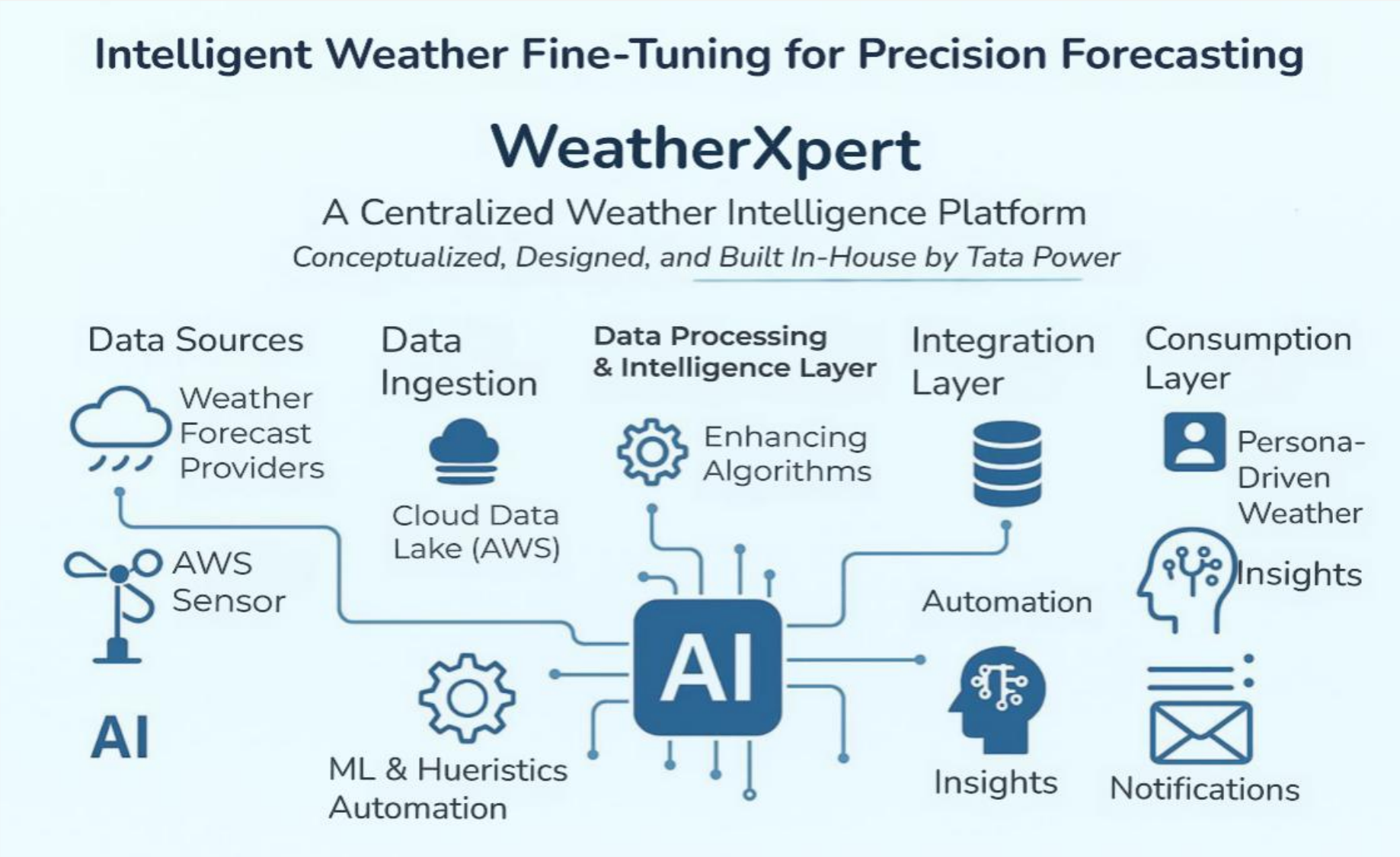
Rule Based Alerting System

Transformer-based language models analyze structured forecast inputs—such as wind speed thresholds, gust probability, timing windows, and asset operating limits—and translate them into clear, operational recommendations for control room teams. By correlating predicted wind gusts with predefined cut-off thresholds and turbine protection parameters, the system can automatically generate precise, time-bound advisories.





RELEVANCE



Built By Utility, For Utilities

Data & AI Powered HYPERLOCAL Weather Intelligence

Our Features

- **One intelligence, driving insights and actions across business functions**
- **Proactive reliability management**
- **Operational Ready by design**
- **Predict > Alert to Prevent or Prepare > Perform across PAN India**

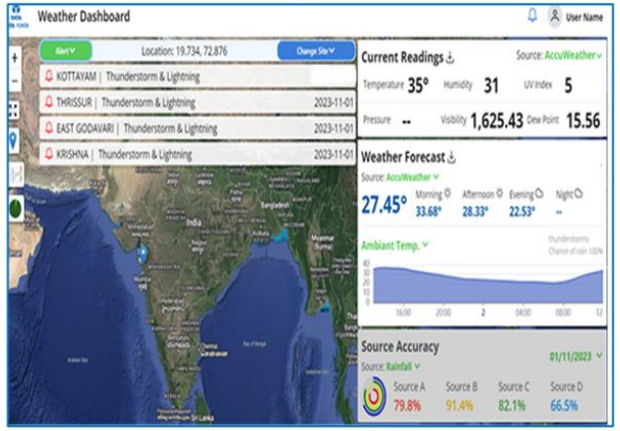
USE CASE / CASE STUDY

India SMART UTILITY Week 2026 | ISGF India Smart Grid Forum

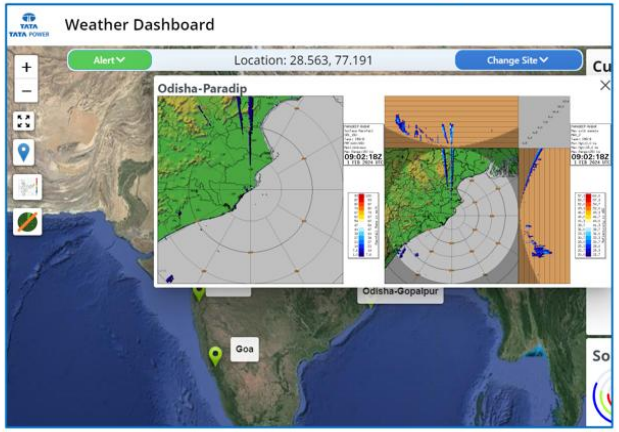
"India changes every 100 km. Now your Weather Platform does too !"

Solution Glimpse

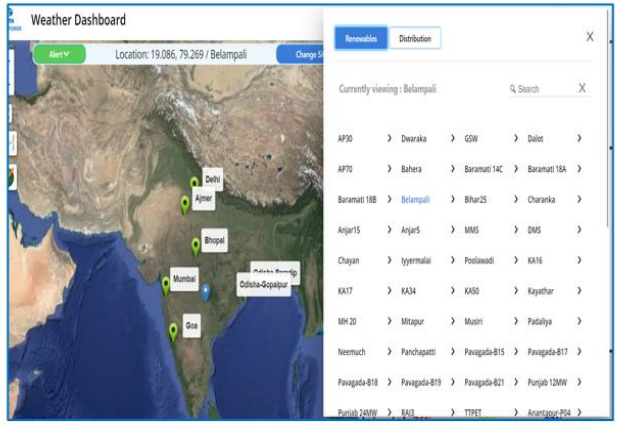
Monitoring & Alerts



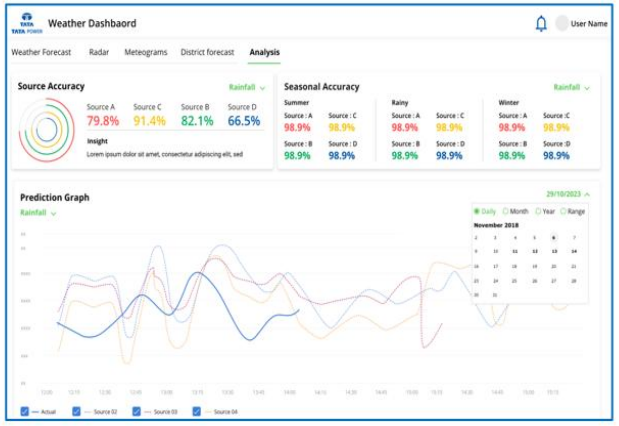
Radar Streaming



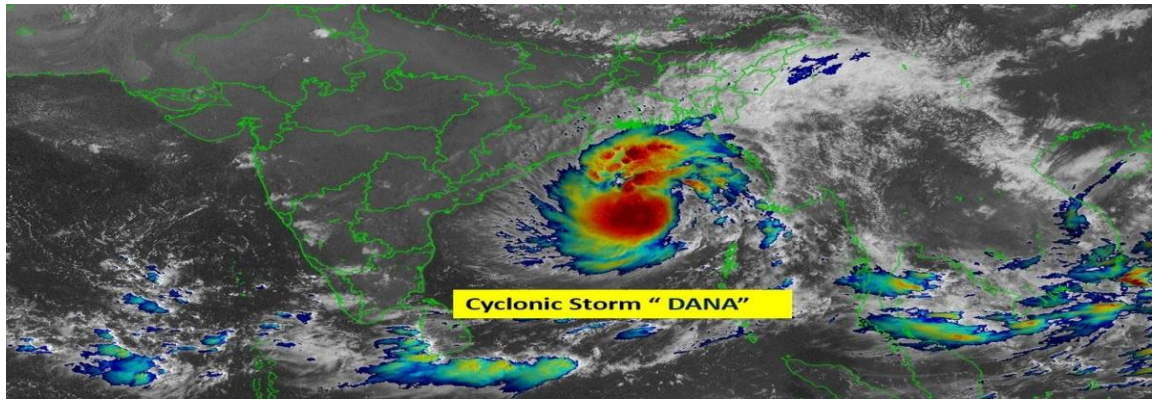
Location Scale (200+)



Accuracy Insights to drive decisions



"Cyclone DANA Case study of WeatherXpert"



WeatherXpert supported Odisha DISCOMs during Cyclone Dana by providing high-frequency live updates on wind speed, rainfall intensity, and storm movement. These location-specific forecasts and impact-based alerts enabled control rooms to proactively shut down vulnerable feeders, protect critical infrastructure, and prevent major equipment damage.

KPI Category	Before WeatherXpert	After WeatherXpert	Improvement
Weather Forecast Accuracy	65%	**94.8%**	**+46%**
DSM/ADSM Penalties	High	**Reduced by 25%**	Direct savings
Renewable Scheduling Efficiency	Baseline	**+7% Improvement**	More RE utilization
Customer Reliability Score	Baseline	**+20% Increase**	Service resilience
Annual ROI	N/A	**>130%**	Strong financial impact

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

www.isuw.in

Links/References (If any)

PRESENTATION GUIDELINES

- **Presentation Time: The total time for presentation is 12 mins. You are requested to be present on-time as per your session slot.**
- Presentation Format: All the presenting speakers are required to give a presentation as per the PPT Format. (Maximum 7 Slides). Please note that the additional presentation slides cannot be added in the shared PPT format.